a04_2

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[3]:
```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.cluster import KMeans

%load_ext autoreload
%autoreload 2

from a04_helper import *
from a04_functions import gmm_e, gmm_m, gmm_fit
```

```
The autoreload extension is already loaded. To reload it, use:
  %reload_ext autoreload
```

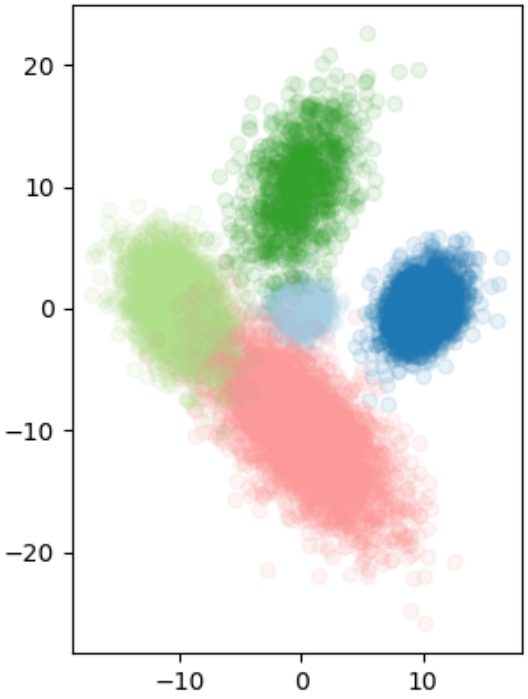
1 2 Gaussian Mixture Models

1.1 2a) Toy data

[4]:
```
# Generate a toy dataset and plot it.
toy_gmm = gmm_gen(
    10000,
    [
        np.array([0, 0]),
        np.array([10, 0]),
        np.array([-10, 0]),
        np.array([0, 10]),
        np.array([0, -10]),
    ],
    np.array([0.1, 0.2, 0.25, 0.1, 0.35]),
    seed=4,
)

print(np.sum(toy_gmm["X"] ** 3))  # must be -4380876.753061278

plot_xy(toy_gmm["X"][:, 0], toy_gmm["X"][:, 1], toy_gmm["components"], alpha=0.
↪1)
```

```
plt.savefig('images/2_toy_ppca', dpi=300, bbox_inches='tight')
```

```
-4380876.753061278
```

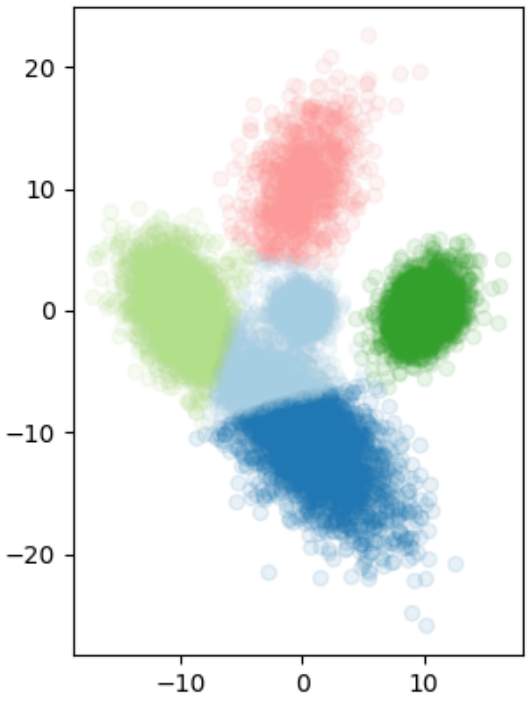


[5]:
```
toy_gmm["X"].shape
```

[5]:
```
(10000, 2)
```

1.2 2b) K-Means

[6]:
```
# Fit 5 clusters using k-means.
kmeans = KMeans(5).fit(toy_gmm["X"])
plot_xy(toy_gmm["X"][:, 0], toy_gmm["X"][:, 1], kmeans.labels_, alpha=0.1)
plt.savefig('images/2_KMeans_toy_gmm', dpi=300, bbox_inches='tight')
```



1.3 2c) Fit a GMM

[7]:
```
# Implement the E step of fitting a GMM with MLE by completing gmm_e in␣
↪a04_functions.py.
```

[8]:
```
# Test your solution. This should produce:
# (array([[9.99999999e-01, 8.65221693e-10, 1.59675131e-23, 1.14015011e-41, 2.
↪78010004e-65],
# [1.00000000e+00, 3.75078862e-12, 1.55035521e-23, 1.28102693e-34, 1.
↪86750812e-46],
# [9.99931809e-01, 6.81161224e-05, 7.23302032e-08, 2.17176125e-09, 1.
↪62736835e-10]]),
# array([[1.71811600e-08, 5.94620494e-18, 1.82893598e-31, 9.79455071e-50, 1.
↪59217812e-73],
# [1.44159783e-15, 2.16285148e-27, 1.48999246e-38, 9.23362817e-50, 8.
↪97398547e-62],
# [1.85095927e-09, 5.04355064e-14, 8.92595932e-17, 2.01005787e-18, 1.
↪00413265e-19]]))
dummy_model = dict(
    mu=[np.array([k, k + 1]) for k in range(5)],
```

```
    Sigma=[np.array([[3, 1], [1, 2]]) / (k + 1) for k in range(5)],
    pi=np.array([0.1, 0.25, 0.15, 0.2, 0.3]),
)
gmm_e(toy_gmm["X"][:3,], dummy_model, return_P=True)
```

[8]:
```
(array([[9.99999999e-01, 8.65221693e-10, 1.59675131e-23, 1.14015011e-41,
         2.78010004e-65],
        [1.00000000e+00, 3.75078862e-12, 1.55035521e-23, 1.28102693e-34,
         1.86750812e-46],
        [9.99931809e-01, 6.81161224e-05, 7.23302032e-08, 2.17176125e-09,
         1.62736835e-10]]),
 array([[1.71811600e-08, 5.94620494e-18, 1.82893598e-31, 9.79455071e-50,
         1.59217812e-73],
        [1.44159783e-15, 2.16285148e-27, 1.48999246e-38, 9.23362817e-50,
         8.97398547e-62],
        [1.85095927e-09, 5.04355064e-14, 8.92595932e-17, 2.01005787e-18,
         1.00413265e-19]]))
```

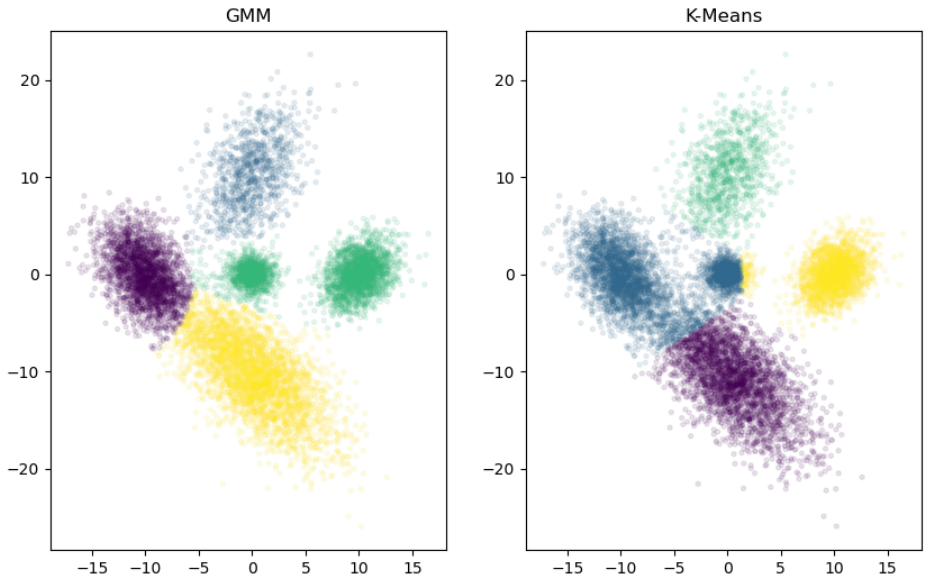
[9]:
```
# Now, implement the M step of fitting a GMM with MLE by completing gmm_e in␣
 ↪a04_functions.py.
```

[10]:
```
# Test your solution. This should produce:
# {'mu': [array([ 6.70641574, -0.47971125]),
#   array([8.2353509 , 2.52134815]),
#   array([-3.0476848 , -1.70722161])],
#  'Sigma': [array([[88.09488652, 11.08635139],
#          [11.08635139,  1.39516823]]),
#   array([[45.82761873, 11.38773232],
#          [11.38773232,  6.87352224]]),
#   array([[98.6662729 , 12.41671355],
#          [12.41671355,  1.56258842]])],
#  'pi': array([0.13333333, 0.53333333, 0.33333333])}
gmm_m(toy_gmm["X"][:3,], np.array([[0.1, 0.2, 0.7], [0.3, 0.4, 0.3], [0.0, 1.0,␣
 ↪0.0]]))
```

[10]:
```
{'mu': array([[ 6.70641574, -0.47971125],
        [ 8.2353509 ,  2.52134815],
        [-3.0476848 , -1.70722161]]),
 'Sigma': array([[[88.09488652, 11.08635139],
         [11.08635139,  1.39516823]],

        [[45.82761873, 11.38773232],
         [11.38773232,  6.87352224]],

        [[98.6662729 , 12.41671355],
         [12.41671355,  1.56258842]]]),
 'pi': array([0.13333333, 0.53333333, 0.33333333])}
```

1.4 2d+2e) Experiment with GMMs for the toy data

[11]:
```
# Fit on toy data and color each point by most likely component. Also try␣
 ↪fitting with 4
# or 6 components.
n_components = 4
toy_gmm_fit = gmm_fit(toy_gmm["X"], n_components)

mu = toy_gmm_fit['mu']
Sigma = toy_gmm_fit['Sigma']
pi = toy_gmm_fit['pi']

W = gmm_e(toy_gmm["X"], toy_gmm_fit)
cluster_labels = np.argmax(W, axis=1)

fig, axes = plt.subplots(1, 2, figsize=(10, 6))

axes[0].set_title('GMM')
axes[0].scatter(toy_gmm["X"][:, 0],
                toy_gmm["X"][:, 1],
                c=cluster_labels,
                s=8, alpha=0.1)
# Kmeans
kmeans = KMeans(n_components).fit(toy_gmm["X"])
axes[1].set_title('K-Means')
axes[1].scatter(toy_gmm["X"][:, 0],
                toy_gmm["X"][:, 1],
                c=kmeans.labels_,
                s=8, alpha=0.1)


plt.show()
plt.savefig(f'images/{n_components}_components_plot_v1.png')
```



[ ]:

[12]:
```
toy_gmm_fit
```

[12]:
```
{'mu': array([[-10.05890454,  -0.01143443],
        [ -0.05319352,  10.24894316],
        [  6.23140675,  -0.02948659],
        [  0.05262685, -10.14139112]]),
 'Sigma': array([[[ 4.05254604, -2.3754577 ],
         [-2.3754577 ,  7.46617118]],

        [[ 5.4685951 ,  3.65771368],
         [ 3.65771368, 13.10737982]],

        [[28.10656618,  0.98292033],
         [ 0.98292033,  2.76897182]],

        [[13.12744617, -9.44523344],
         [-9.44523344, 15.68224926]]]),
 'pi': array([0.24049833, 0.09798413, 0.31676301, 0.34475453])}
```

[13]:
```
toy_gmm['X'].shape
```

[13]:
```
(10000, 2)
```

1.5 2f) Discover the Secret (optional)

[14]:
```
# Load the secret data.
X = np.loadtxt("data/secret_gmm.csv", delimiter=",")
```